Q1 2021 (27 Aug Shift 2)

Let $\mathrm{A}(\sec \theta, 2 \tan \theta)$ and $\mathrm{B}(\sec \phi, 2 \tan \phi)$, where $\theta+\phi=\pi / 2$, be two points on the hyperbola $2 \mathrm{x}^2-\mathrm{y}^2=2$. If (α, β) is the point of the intersection of the normals to the hyperbola at A and B, then $(2 \beta)^2$ is equal to .

Q2 2021 (26 Aug Shift 2)

The locus of the mid points of the chords of the hyperbola $x^2-y^2=4$, which touch the parabola $\mathrm{y}^2=8 \mathrm{x}$, is :

(1) $\mathrm{y}^3(\mathrm{x}-2)=\mathrm{x}^2$

(2) $x^3(x-2)=y^2$

(3) $y^2(x-2)=x^3$

(4) $x^2(x-2)=y^3$

Q3 2021 (26 Aug Shift 2)

The point $\mathrm{P}(-2 \sqrt{6}, \sqrt{3})$ lies on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ having eccentricity $\frac{\sqrt{5}}{2}$. If the tangent and normal at P to the hyperbola intersect its conjugate axis at the point Q and R respectively, then QR is equal to :

(1) $4 \sqrt{3}$

(2) 6

(3) $6 \sqrt{3}$

(4) $3 \sqrt{6}$

Answer Key

Q1 (36) **Q2** (3) **Q3** (3)

Q1 (36)

Since, point $A(\sec\theta, 2\tan\theta)$

lies on the hyperbola

$2x^2 - y^2 = 2$

Therefore, $2\sec^2\theta - 4\tan^2\theta = 2$

$\Rightarrow 2 + 2\tan^2\theta - 4\tan^2\theta = 2$

$\Rightarrow \tan\theta = 0 \Rightarrow \theta = 0$

Similarly, for point B, we will get $\phi = 0$.

but according to question $\theta + \phi = \frac{\pi}{2}$

which is not possible.

Hence it must be a $'BONUS'$.

Q2 (3)

$\mathrm{T} = \mathrm{S}_1$

$\mathrm{xh} - \mathrm{yk} = \mathrm{h}^2 - \mathrm{k}^2$

$\mathrm{y} = \frac{\mathrm{xh}}{\mathrm{k}} - \frac{\left(\mathrm{h}^2 - \mathrm{k}^2\right)}{\mathrm{k}}$

this touches $y^2 = 8x$ then $c = \frac{a}{m}$

$\left(\frac{\mathrm{k}^2 - \mathrm{h}^2}{\mathrm{k}}\right) = \frac{2\mathrm{k}}{\mathrm{h}}$

$2y^2 = x\left(y^2 - x^2\right)$

$y^2(x-2) = x^3$

Q3 (3)

$\mathrm{P}(-2\sqrt{6}, \sqrt{3})$ lies on hyperbola

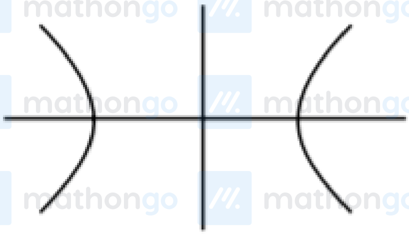
$\Rightarrow \frac{24}{a^2} - \frac{3}{b^2} = 1$........(i)

$e = \frac{\sqrt{5}}{2} \Rightarrow b^2 = a^2\left(\frac{5}{4} - 1\right) \Rightarrow 4b^2 = a^2$

Put in (i) $\Rightarrow \frac{6}{\mathrm{b}^2} - \frac{3}{\mathrm{b}^2} = 1 \Rightarrow \mathrm{b} = \sqrt{3}$

$\Rightarrow \mathrm{a} = \sqrt{12}$

$\frac{x^2}{12} - \frac{y^2}{3} = 1$



Tangent at P :

$\frac{-\mathrm{x}}{\sqrt{6}}-\frac{\mathrm{y}}{\sqrt{3}}=1 \Rightarrow \mathrm{Q}(0, \sqrt{3})$

Slope of $\mathrm{T}=-\frac{1}{\sqrt{2}}$

Normal at P :

$y-\sqrt{3}=\sqrt{2}(x+2 \sqrt{6})$

$\Rightarrow \mathrm{R}=(0,5 \sqrt{3})$

$\mathrm{QR}=6 \sqrt{3}$